



ENGINEERING MATHEMATICS-II

UE25MA141B

Department of Science and Humanities



ENGINEERING MATHEMATICS-II

UNIT 1: INTEGRAL CALCULUS

class 1: DOUBLE INTEGRAL: INTRODUCTION & EVALUATION

TOPICS COVERED

1. DOUBLE INTEGRAL: INTRODUCTION and evaluation
2. Application of double integral: area, volume of the solid and Average value of the function.
3. Jacobian, CHANGE OF VARIABLES IN DOUBLE INTEGRAL
4. Problems on change of variables
5. CHANGE OF ORDER OF INTEGRATION
6. TRIPLE INTEGRAL: Introduction and evaluation
7. Application of triple integral: volume
8. Finding average value using triple integral
9. CHANGE OF VARIABLES IN TRIPLE INTEGRAL: cylindrical coordinates
10. Change of variables in triple integral: spherical coordinates
11. APPLICATIONS: CENTER OF MASS
12. APPLICATIONS: MOMENT OF INERTIA



INTRODUCTION TO DOUBLE INTEGRAL



The definite integral can be extended to functions of more than one variable. Consider, for example, a function of two variables $z = f(x, y)$. The double integral of function $f(x, y)$ is denoted by

$$\iint_R f(x, y) dA,$$

where R is the region of integration in the xy -plane.

If the definite integral $\int_a^b f(x) dx$ of a function of one variable $f(x) \geq 0$ is the area under the curve $f(x)$ from $x = a$ to $x = b$, then the double integral is equal to the volume under the surface $z = f(x, y)$ and above the xy -plane in the region of integration R (Figure 1).

DOUBLE INTEGRAL DEFINITION

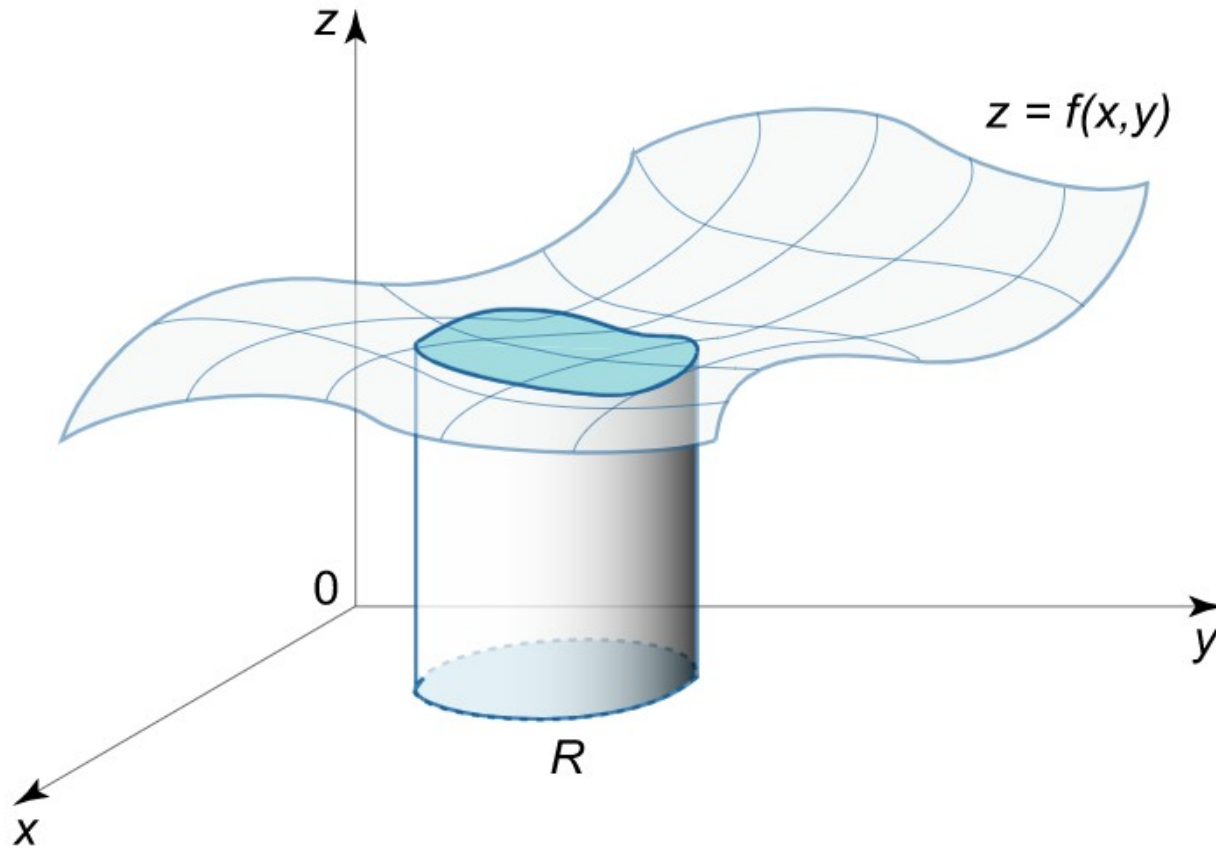


Figure 1.

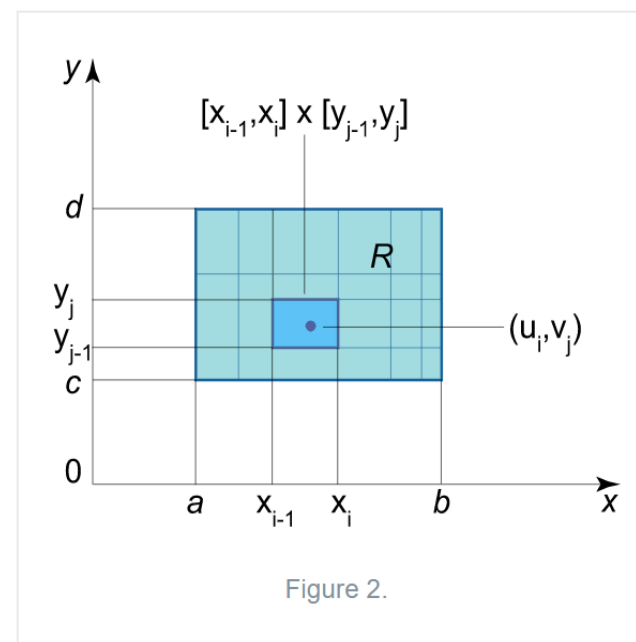
DOUBLE INTEGRAL DEFINITION CONTINUED...



As in the case of integral of a function of one variable, a double integral is defined as a limit of a Riemann sum.

If the region R is a rectangle $[a, b] \times [c, d]$ (Figure 2), we can subdivide $[a, b]$ into small intervals with a set of numbers $\{x_0, x_1, \dots, x_m\}$ so that

$$a = x_0 < x_1 < x_2 < \dots < x_i < \dots < x_{m-1} < x_m = b.$$



INTERPRETATION OF DOUBLE INTEGRAL



Similarly, a set of numbers $\{y_0, y_1, \dots, y_n\}$ is said to be a partition of $[c, d]$ along the y -axis, if

$$c = y_0 < y_1 < y_2 < \dots < y_j < \dots < y_{n-1} < y_n = d.$$

The **Riemann sum** of a function $f(x, y)$ over this partition of $[a, b] \times [c, d]$ is

$$\sum_{i=1}^m \sum_{j=1}^n f(u_i, v_j) \Delta x_i \Delta y_j,$$

where (u_i, v_j) is some point in the rectangle $(x_{i-1}, x_i) \times (y_{j-1}, y_j)$ and $\Delta x_i = x_i - x_{i-1}$,

$$\Delta y_j = y_j - y_{j-1}.$$

DEFINITION OF DOUBLE INTEGRAL



Definition. We call the **double integral** of $f(x, y)$ under the region R the value

$$\iint_R f(x, y) \, dA = \lim_{n \rightarrow +\infty} \sum_{k=1}^n f(x_k^*, y_k^*) \Delta A_k.$$

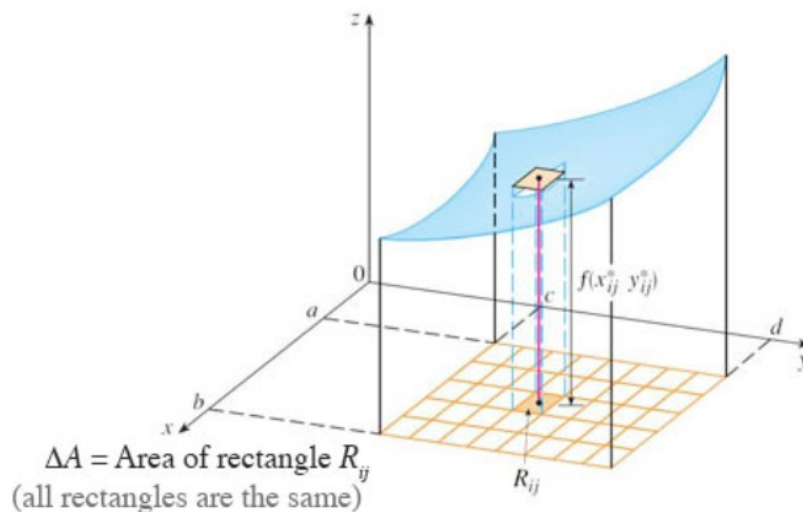
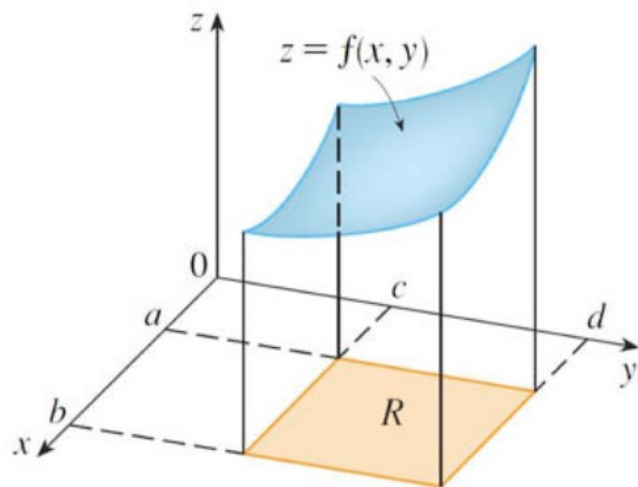
Therefore, the volume V that we intended to determine is given by

$$V = \iint_R f(x, y) \, dA = \iint_R f(x, y) \, dx \, dy.$$

FUBINI'S THEOREM FOR RECTANGULAR REGION

Let R be the rectangle defined by $a \leq x \leq b$, $c \leq y \leq d$. If $f(x, y)$ is continuous on that rectangle, then

$$\iint_R f(x, y) dA = \int_c^d \int_a^b f(x, y) dx dy = \int_a^b \int_c^d f(x, y) dy dx$$



FUBINI'S THEOREM FOR RECTANGULAR REGION



Use Fubini's theorem to compute the double integral $\iint_R f(x, y) dA$ where $f(x, y) = x$ and $R = [0, 2] \times [0, 1]$.

Solution

Fubini's theorem offers an easier way to evaluate the double integral by the use of an iterated integral. Note how the boundary values of the region R become the upper and lower limits of integration.

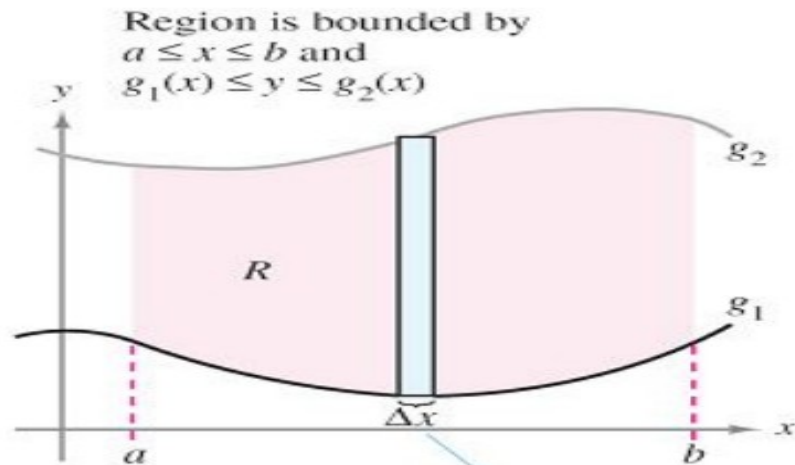
$$\begin{aligned}\iint_R f(x, y) dA &= \iint_R f(x, y) dx dy \\&= \int_{y=0}^{y=1} \int_{x=0}^{x=2} x dx dy \\&= \int_{y=0}^{y=1} \left[\frac{x^2}{2} \right]_{x=0}^{x=2} dy \\&= \int_{y=0}^{y=1} 2 dy = 2y \Big|_{y=0}^{y=1} = 2\end{aligned}$$

FUBINI'S THEOREM FOR NON RECTANGULAR REGION

If R is not a rectangular region, then we have to distinguish between two types of regions: • **Regions type I**

$$R = \left\{ (x, y) \in \mathbb{R}^2 : a \leq x \leq b, g_1(x) \leq y \leq g_2(x) \right\}$$

such that $g_1(x)$ and $g_2(x)$ are continuous in $[a, b]$.



$$\text{Area} = \int_a^b \int_{g_1(x)}^{g_2(x)} dy \, (dx)$$

Vertically simple region

FUBINI'S THEOREM FOR NON RECTANGULAR REGIONS

- **Regions type II**

$$R = \left\{ (x, y) \in \mathbb{R}^2 : h_1(y) \leq x \leq h_2(y), c \leq y \leq d \right\}$$

such that $h_1(y)$ and $h_2(y)$ are continuous in $[c, d]$.

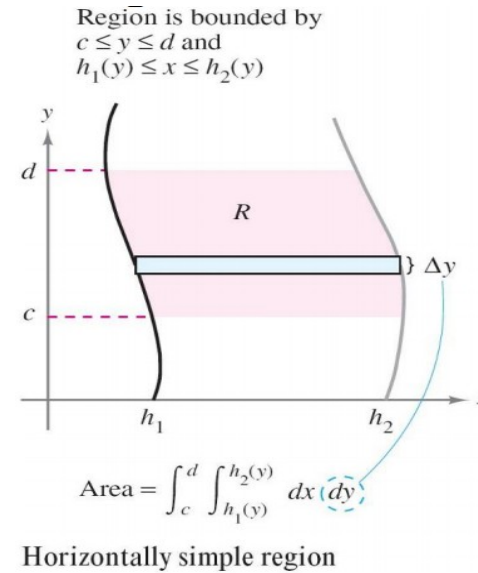
Let $f(x, y)$ be continuous in the region R .

- *If R is a region of type I, then*

$$\iint_R f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx.$$

- *If R is a region of type II, then*

$$\iint_R f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy.$$



PROPERTIES OF DOUBLE INTEGRALS



Let f and g be continuous over a closed, bounded plane region R , and let c be a constant.

$$1. \int_R \int cf(x, y) dA = c \int_R \int f(x, y) dA$$

$$2. \int_R \int [f(x, y) \pm g(x, y)] dA = \int_R \int f(x, y) dA \pm \int_R \int g(x, y) dA$$

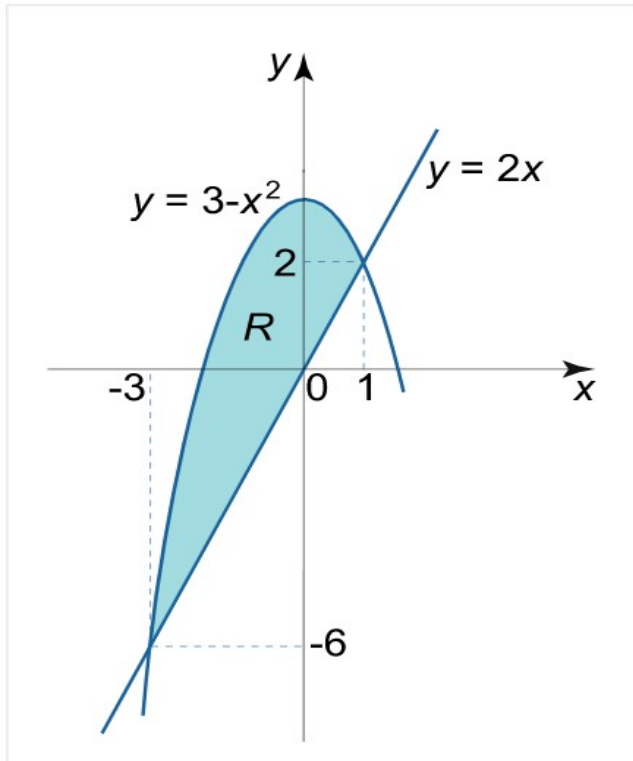
$$3. \int_R \int f(x, y) dA \geq 0, \quad \text{if } f(x, y) \geq 0$$

$$4. \int_R \int f(x, y) dA \geq \int_R \int g(x, y) dA, \quad \text{if } f(x, y) \geq g(x, y)$$

$$5. \int_R \int f(x, y) dA = \int_{R_1} \int f(x, y) dA + \int_{R_2} \int f(x, y) dA, \text{ where } R \text{ is the union of two nonoverlapping subregions } R_1 \text{ and } R_2.$$

PROPERTIES OF DOUBLE INTEGRALS

Find the integral $\iint_R y dy dx$, where R is bounded by the straight line $y = 2x$ and parabola $y = 3 - x^2$.

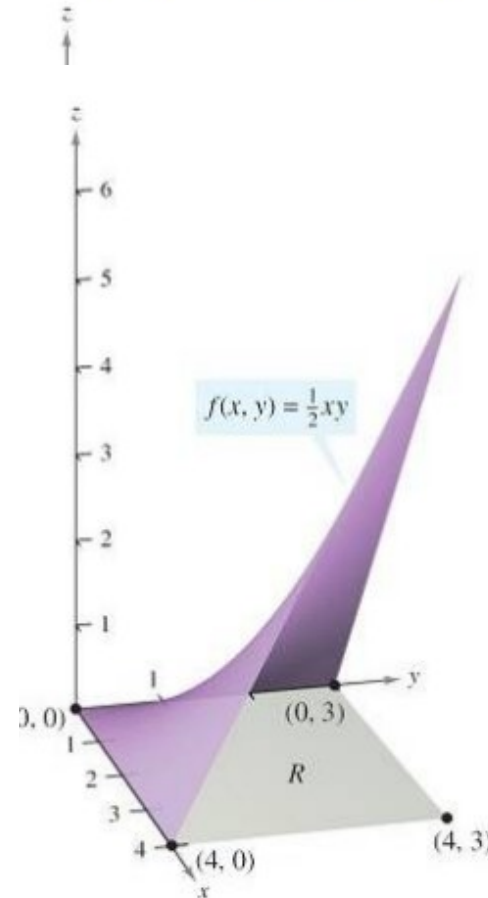


PROPERTIES OF DOUBLE INTEGRALS

Find the average value of $f(x, y) = \frac{1}{2}xy$ over the region R , where R is a rectangle with vertices $(0, 0)$, $(4, 0)$, $(4, 3)$, and $(0, 3)$.

The average value is given by

$$\begin{aligned}\frac{1}{A} \iint_R f(x, y) \, dA &= \frac{1}{12} \int_0^4 \int_0^3 \frac{1}{2}xy \, dy \, dx \\&= \frac{1}{12} \int_0^4 \left[\frac{1}{4}xy^2 \right]_0^3 \, dx \\&= \left(\frac{1}{12} \right) \left(\frac{9}{4} \right) \int_0^4 x \, dx \\&= \frac{3}{16} \left[\frac{1}{2}x^2 \right]_0^4 \\&= \left(\frac{3}{16} \right) (8) \\&= \frac{3}{2}.\end{aligned}$$





THANK YOU
